

NOTE ON THE CONSTRUCTION OF DISTRIBUTION-FREE TESTS FOR RANDOMISED BLOCK DESIGNS

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1. Introduction and summary

Test statistics for treatment effects in randomised block designs are constructed by using different functions of ranks in the different blocks. The functions are chosen to maximise the a.r.e.'s of the tests (with respect to the parametric F test) — cf. Sen (1968) who used the same function in the different blocks.

The test statistics obtained have high relative efficiencies with respect to the F test, especially when the variances σ_s^2 of the underlying distributions with c.d.f.'s F_s differ.

The class of tests considered follows as a special case from Lemmer, Stoker and Reinach (1968), when

$$(1) \quad k_{1j} = k_{2j} = \dots = k_{mj} = b_j \text{ for each } j=1, \dots, k.$$

The reader is referred to Lemmer, Stoker and Reinach (1968) for the notation being used here.

2. Main results

Under condition (1) and if

$$(2) \quad \sum_j b_j d_j = 0,$$

the test statistic T of Lemmer, Stoker and Reinach (1968) reduces to

$$(3) \quad T = \frac{a-1}{a\sigma_\Psi^2} \sum_j t_j^2 / b_j$$

where $a = \sum_j b_j$, $\sigma_\Psi^2 = (am)^{-1} \sum_s \sum_{p=1}^a (\Psi_{sp} - \bar{\Psi}_s)^2$, $\Psi_{sp} = \Psi_s(\frac{p}{a+1})$ and

$$t_j = m^{-\frac{1}{2}} \sum_s \sum_{h=1}^{b_j} (\Psi_{sjh} - \bar{\Psi}_s).$$

From theorem 1 of Lemmer, Stoker and Reinach (1968) we have: If all $F_s(x)$ have continuous and uniformly bounded frequency functions $f_s(x)$, then T has under H asymptotically, for $m \rightarrow \infty$, a non-central χ^2 distribution with $(k-1)$ degrees of freedom and non-centrality parameter

$$\lambda^2 = a(a-1)\sigma_\Psi^2 \sum_j b_j d_j^2 (m^{-1} \sum_s \sum_p \Psi_{sp} t_{p,s})^2$$

where

$$(4) \quad t_{p,s} = \binom{a-2}{p-2} \int_{-\infty}^{\infty} F_s^{p-2}(x) [1-F_s(x)]^{a-p} f_s^2(x) dx \\ - \binom{a-2}{p-1} \int_{-\infty}^{\infty} F_s^{p-1}(x) [1-F_s(x)]^{a-p-1} f_s^2(x) dx$$

for $1 \leq p \leq a$, $1 \leq s \leq m$ with $\binom{n}{r} = 0$ for $r > n$ or $r < 0$. (Cf. Lemmer, Stoker and Reinach (1968) p. 15).

Let $X_{p,a}^{(s)}$ be the p -th order statistic of a random sample of size a from the distribution with c.d.f. F_s , then

$$(5) \quad E[f_s(X_{p,a-1}^{(s)})] = (a-1) \binom{a-2}{p-1} \int_{-\infty}^{\infty} F_s^{p-1}(x) [1-F_s(x)]^{a-p-1} f_s^2(x) dx.$$

From (4) $t_{p,s}$ reduces to

$$(6) \quad t_{p,s} = (a-1)^{-1} \{E[f_s(X_{p-1,a-1}^{(s)})] - E[f_s(X_{p,a-1}^{(s)})]\}$$

for $1 \leq p \leq a$, $1 \leq s \leq m$ if we define $E[f_s(X_{0,a-1}^{(s)})] = 0$ and $E[f_s(X_{a,a-1}^{(s)})] = 0$.

The non-centrality parameter of the parametric F test is

$$\lambda_F^2 = \sum_j b_j d_j^2 / \bar{\sigma}^2 \text{ where } \bar{\sigma}^2 = m^{-1} \sum_s \sigma_s^2$$

The a.r.e. of T with respect to the F test thus becomes

$$E_{T,F} = \lambda^2 / \lambda_F^2 = a(a-1) \bar{\sigma}^2 \sigma_{\Psi}^2 [m^{-1} \sum_s \sum_p \Psi_{sp} t_{p,s}]^2 \\ = a^2 (a-1) \bar{\sigma}^2 m^{-1} \sum_s \sum_p t_{p,s}^2$$

if we take

$$(7) \quad \Psi_{sp} = \bar{\Psi}_s = \text{const. } t_{p,s} \text{ for } 1 \leq p \leq a \text{ and } 1 \leq s \leq m,$$

a substitution that maximises $E_{T,F}$ (cf. Sen (1968)).

3. Application: The normal distribution

If f_s and f are the frequency functions of the $N(\mu_s, \sigma_s^2)$ and $N(0,1)$ distributions respectively and $Z_{p,a}$ the p -th order statistic of a random sample of size a from the $N(0,1)$ distribution, then

$$(8) \quad E[f_s(X_{p,a-1}^{(s)})] = \sigma_s^{-1} E[f(Z_{p,a-1})].$$

From (5) and equation (3.12) of Terry (1952) we have:

$$(9) \quad E[f(Z_{p,a-1})] = a^{-1} \sum_{u=p+1}^a E(Z_{u,a}).$$

Substitution of (8) and (9) in (6) gives

$$t_{p,s} = E(Z_{p,a})/a(a-1)\sigma_s.$$

The most efficient test according to the above method, is obtained (cf. equation (7)) by taking

$$\Psi_{sp} = \sigma_s^{-1} E(Z_{p,a}) \text{ for } 1 \leq p \leq a, 1 \leq s \leq m.$$

The test statistic is

$$T_N = \frac{(a-1) \sum_j b_j^{-1} \left[\sum_s \sigma_s^{-1} \sum_h E(Z_{r_{sjh},a}) \right]^2}{\sum_s \sigma_s^2 \sum_p [E(Z_{p,a})]^2}$$

with a.r.e.

$$E_{T_N, F} = B(a-1)^{-1} \sum_p [E(Z_{p,a})]^2$$

$$\text{where } B = (m^{-1} \sum_s \sigma_s^2)(m^{-1} \sum_s \sigma_s^{-2}) \geq 1$$

a factor that increases rapidly the more the variances σ_s^2 , $s = 1, \dots, m$ differ. Herein lies the strength of this method.

Extensive tables of $E(Z_{p,a})$ are available in Harter (1960).

A few a.r.e. values are shown below.

a :	4	6	8	10	12	20
$E_{T_N, F}$:	0.7652B	0.8233B	0.8571B	0.8794B	0.8953B	0.9304B

4. Remarks

- (1) A very important aspect from a practical point of view is that the test statistics deduced from T and their asymptotic relative efficiencies can be calculated even if the variances σ_s^2 are not known explicitly, provided only that their ratios are known.
- (2) The functions F_s need not necessarily be all of the same type in a specific application.
- (3) A simplification of expression (39) in Lemmer, Stoker and Reinach (1968) in case of normality is easily obtained, viz.

$$E_{T_{c,F}} = [kc(kc-1)\sigma^2]^{-1} \left[\sum_p Q\left(\frac{p}{kc+1}\right) E(Z_{p,kc}) \right]^2.$$

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